

Each question is worth 5 points for a total of 30 points. The extra Credit can get you an additional 5 points making the highest possible score 35 out of 30. Write solutions using COMPLETE sentences.

Problem 1

Prove by contradiction: If the sum of two primes is prime, then one of the primes must be 2. You may assume that every integer is either even or odd, but never both.

Problem 2

Prove that $1 + 2 + 3 + \dots + n = \frac{1}{2}n(n + 1)$ for all positive integers n . You may use any proof technique you prefer.

Problem 3

Prove $n! \leq n^n$ for all positive integer n . You may use any proof technique you prefer.

Problem 4

Prove $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$ for all positive integer n . You may use any proof technique you prefer.

Problem 5

Prove $2^n \leq 2^{n+1} - 2^{n-1} - 1$ for all positive integer n . You may use any proof technique you prefer.

Problem 6

Prove that the equation $x^2 + 1 = 0$ does not have any real solutions.

Extra Credit:

This problem is one that you have until next to complete for extra credit. Prove that for all natural numbers n , then

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

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